

5.3.4

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Question

Find the shortest distance between the lines:

$$\frac{x-1}{2} = \frac{y+1}{3} = z, \quad \frac{x+1}{5} = \frac{y-2}{1}, \quad z = 2$$

Solution:

The vector equations of the lines are

$$\mathbf{r}_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + k_1 \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} \quad (1)$$

$$\mathbf{r}_2 = \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix} + k_2 \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix} \quad (2)$$

Let

$$\mathbf{A} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix}, \quad \mathbf{m}_1 = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \quad \mathbf{m}_2 = \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix} \quad (3)$$

The shortest distance between two skew lines is obtained by solving

$$(M^T M)\mathbf{K} = M^T(\mathbf{B} - \mathbf{A}) \quad (4)$$

where

$$M = (\mathbf{m}_1 \quad -\mathbf{m}_2) = \begin{pmatrix} 2 & -5 \\ 3 & -1 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{B} - \mathbf{A} = \begin{pmatrix} -2 \\ 3 \\ 2 \end{pmatrix} \quad (5)$$

Computing,

$$M^T M = \begin{pmatrix} 14 & -13 \\ -13 & 26 \end{pmatrix}, \quad M^T(\mathbf{B} - \mathbf{A}) = \begin{pmatrix} 7 \\ 7 \end{pmatrix} \quad (6)$$

Hence,

$$\begin{pmatrix} 14 & -13 \\ -13 & 26 \end{pmatrix} \mathbf{K} = \begin{pmatrix} 7 \\ 7 \end{pmatrix} \quad (7)$$

Solving,

$$\mathbf{K} = \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} \frac{7}{5} \\ \frac{63}{65} \end{pmatrix} \quad (8)$$

Substitute to find the points on the lines:

$$\mathbf{x}_1 = \mathbf{A} + k_1 \mathbf{m}_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + \frac{7}{5} \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{19}{5} \\ \frac{16}{5} \\ \frac{7}{5} \end{pmatrix} \quad (9)$$

$$\mathbf{x}_2 = \mathbf{B} + k_2 \mathbf{m}_2 = \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix} + \frac{63}{65} \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{50}{13} \\ \frac{193}{65} \\ 2 \end{pmatrix} \quad (10)$$

Therefore, the minimum distance is

$$d = \|\mathbf{x}_1 - \mathbf{x}_2\| = \left\| \begin{pmatrix} \frac{19}{5} - \frac{50}{13} \\ \frac{16}{5} - \frac{193}{65} \\ \frac{7}{5} - 2 \end{pmatrix} \right\| = \frac{9}{\sqrt{195}} \quad (11)$$

Minimum Distance between the given Lines = $\frac{9}{\sqrt{195}}$ units
